Jacob E. Brinton

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EDUCATION

Boston University *Expected May 2026*
B.A./M.S. in Mathematics and Computer Science Major GPA: 4.0

- **Relevant coursework:** Interpretable Machine Learning, Directed Study in LLM Translation Interpretability, Principles of ML, Natural Language Processing, Stochastic Processes, Causal Inference*, Abstract Algebra, Advanced Algorithms, Intro to AI, Foundations of Mathematics*, Logic and Programming Languages*, Discrete Mathematics (* in progress)

Utah State University *Aug 2022 – May 2023*
GPA: 4.0

- **Relevant coursework:** Honors Calculus II & III, Chemistry, Physics, Programming in C, CAD

RESEARCH AND PROFESSIONAL EXPERIENCE

Executive Director *May 2025 – Present*
Boston University Artificial Intelligence Safety Association Boston, MA

- Grew club from 4–6 person avg. attendance to 120 total applicants across 6 intro cohorts and 2 upskilling programs.
- Delivered tailored presentations on AI safety to over a dozen lectures, reaching a total enrollment of 1,000+ students.
- Organized guest speakers and intra-university partnerships, putting on events with 90+ attendees.
- Developed technical AI safety curriculum and presentations on goal misgeneralization, interpretability, and evaluations.

Research Fellow *Dec 2025 – Present*
Mentorship for Alignment Researchers (CAISH) Cambridge, UK

- Developing tamper-resistance methods that constrain harmful-capability transfer across behavior categories, with an RL training component; mentored by Prof. Zhijing Jin.

Teaching Fellow, Course Assistant *Aug 2024 – Present*
Boston University College of Arts and Sciences Boston, MA

- Guided labs on computer architecture, operating systems, memory management, concurrency, and networking.
- Led office hours, debugging issues in Jade circuitry, x86 Assembly, Git, and C, along with calculus and linear algebra.

Machine Learning Research Intern *May 2023 – Aug 2023*
Montana State University NSF REU Bozeman, MT

- Designed and conducted drone-flight field experiments, gathering over 3 hours of audio.
- Cleaned 1M audio frames in MATLAB; used scikit-learn for clustering, dimension reduction, and signal processing.

Student Researcher *Dec 2022 – May 2023*
Direct Laboratory Logan, UT

- Leveraged ROS to write Python-based publishers and subscribers for robot pathfinding and camera movement.

Ground Station Subteam Member *Sep 2022 – May 2023*

Get Away Special Satellite Team Logan, UT

- 3D-modeled and manufactured prototypes in SolidWorks for an S-band patch antenna connection and coverglass frame.

PUBLICATIONS

Assessing LWIR and Visual Sensor Modalities for UAV Detection and Tracking *May 2024*

- Coauthored paper assessing long-wave infrared (LWIR) and visual sensor modalities for UAV detection.
- *2024 IEEE Intermountain Engineering, Technology and Computing (IETC), pp. 141–146.* doi:10.1109/IETC61393.2024.10564431

PROJECTS

Noisy Translation Channels in Large Language Models *Mar 2025 – Present*

- Evaluated machine translation using Llama-3.1-8B and Aya-23-8B to interpret effects of English-centric pretraining data.
- Intervened on SAE features relevant to multi- and monolingual grammatical concept representation for causal relevance.
- Quantified multilinguality of features' receptive fields and output spaces, and representation similarity across layers.

Cross-Language Synonym Analysis and Scoring *Sep 2024 – Jan 2025*

- Developed system to quantify closeness of synonyms across languages and their translation flexibility.
- Used Pandas and spaCy for tokenization and lemmatization; validated model on the Tatoeba dataset.

Tetris Reinforcement Learning Agent *Apr 2024 – May 2024*

- Built Tetris agent in Java, integrating a custom neural network library and using teacher forcing to speed training times.
- Earned 2nd place out of 150-student course for feature engineering and effective hyperparameter search.

PRESENTATIONS AND POSTERS

Poster: Noisy Translation Channels in Large Language Models *2025*

New England Mechanistic Interpretability workshop (NEMI)

AWARDS AND FELLOWSHIPS

- **Boston University Undergraduate Research Opportunities Program (UROP) — Summer Research Fellowship** *Summer 2025*
  Funded the Noisy Translation Channels project; presented poster at NEMI.
- **Tetris Reinforcement Learning Competition — 2nd of 150** *May 2024*
  CS 506 Tools for Data Science, Boston University.

TECHNICAL SKILLS AND LANGUAGES

Technical: Python, PyTorch, HuggingFace, nnsight, Weights & Biases, Claude Code, OCaml, C.
Languages: English (Native), Spanish (Fluent).